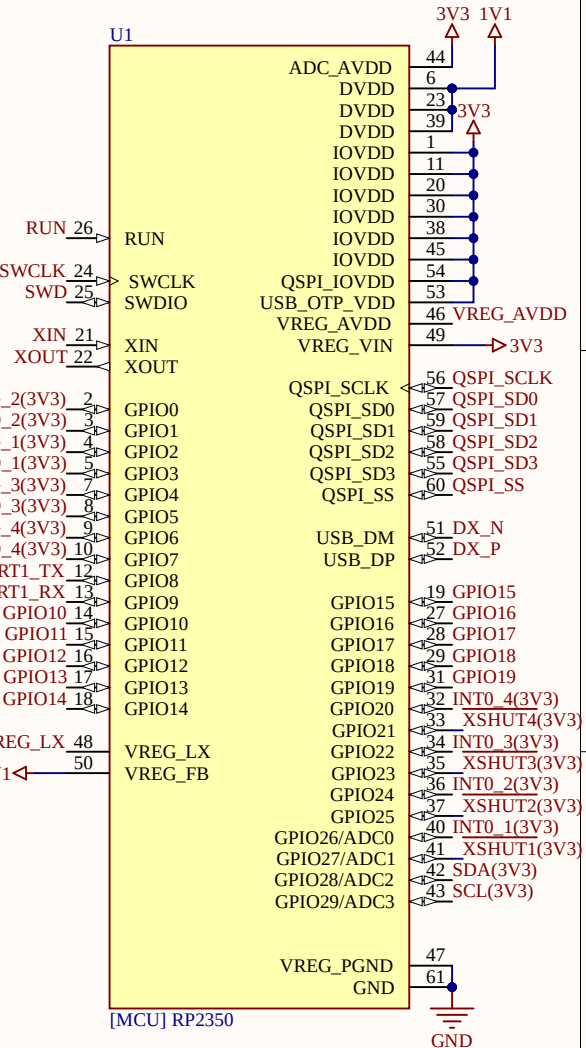
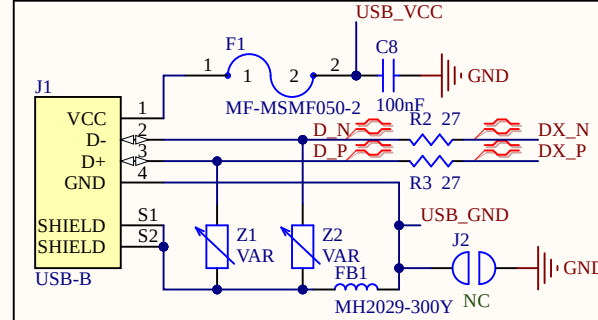


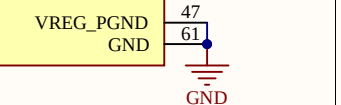
$$\frac{3V_3}{\dot{I}_3} = \frac{1V_1}{\dot{I}_1}$$


Pin	Function	Pin	Function
26	RUN	26	RUN
24	SWCLK	24	SWCLK
25	SWD	25	SWDIO
21	XIN	21	XIN
22	XOUT	22	XOUT
2	TRIG_2(3V3)	2	GPIO0
3	ECHO_2(3V3)	3	GPIO1
4	TRIG_1(3V3)	4	GPIO2
5	ECHO_1(3V3)	5	GPIO3
7	TRIG_3(3V3)	7	GPIO4
8	ECHO_3(3V3)	8	GPIO5
9	TRIG_4(3V3)	9	GPIO6
10	ECHO_4(3V3)	10	GPIO7
12	UART1_TX	12	GPIO8
13	UART1_RX	13	GPIO9
14	GPIO10	14	GPIO10
15	GPIO11	15	GPIO11
16	GPIO12	16	GPIO12
17	GPIO13	17	GPIO13
18	GPIO14	18	GPIO14
48	VREG_LX	48	VREG_L
50	1V1	50	VREG_F

QSPI_SCLK	56 QSPI_SCLK
QSPI_SD0	57 QSPI_SD0
QSPI_SD1	59 QSPI_SD1
QSPI_SD2	58 QSPI_SD2
QSPI_SD3	55 QSPI_SD3
QSPI_SS	60 QSPI_SS
USB_DM	51 DX_N
USB_DP	52 DX_P
GPIO15	19 GPIO15
GPIO16	27 GPIO16
GPIO17	28 GPIO17
GPIO18	29 GPIO18
GPIO19	31 GPIO19
GPIO20	32 INT0_4(3V3)
GPIO21	33 XSHUT4(3V3)
GPIO22	34 INT0_3(3V3)
GPIO23	35 XSHUT3(3V3)
GPIO24	36 INT0_2(3V3)
GPIO25	37 XSHUT2(3V3)
GPIO26/ADC0	40 INT0_1(3V3)
GPIO27/ADC1	41 XSHUT1(3V3)
GPIO28/ADC2	42 SDA(3V3)
GPIO29/ADC3	43 SCL(3V3)



VREG_LX $\frac{48}{50}$ VREG_LX
1V1 $\leftarrow$ VREG_FB

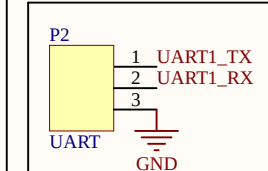


2x10 Extra GPIO Breakout (and debug connections)

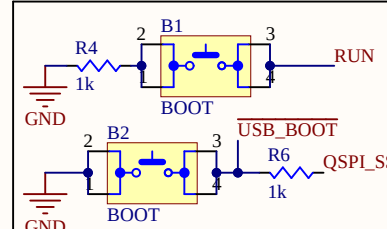
Pin connection diagram for the J3 connector:

GPIO Pin	Connector Pin	Signal
GPIO10	1	
GPIO11	3	
GPIO12	5	
GPIO13	7	
GPIO14	9	
GPIO15	11	
GPIO16	13	
GPIO17	15	
GPIO18	17	
GPIO19	19	
	2	3V3
	4	SWCLK
	6	SWD
	8	SWD
	10	SWD
	12	SWD
	14	SWD
	16	SWD
	18	SWD
	20	GND

[Connector] 879142016



- MIL
- MCU setup according to RPi's hardware design guidelines



Revisor(s):	- *
	- *
	- *
	- *

```
Git Repo: *
Git Hash:
```



A

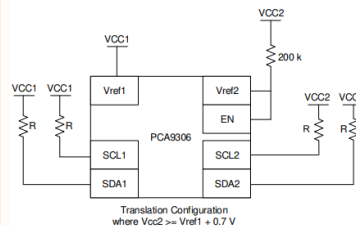
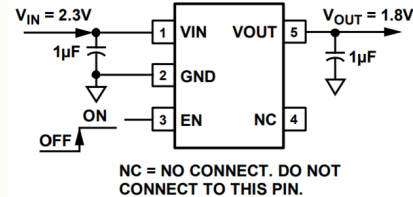
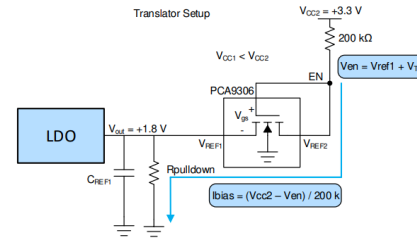


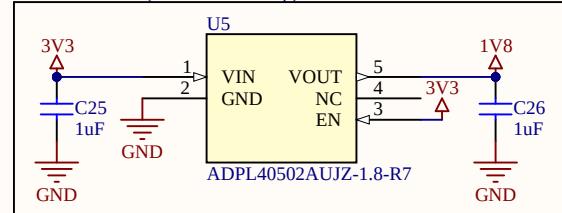
Figure 8-6. Translation Configuration



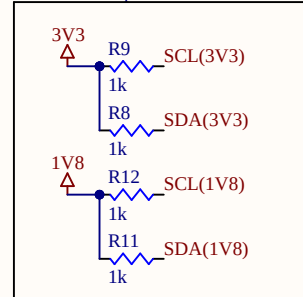
001



3v3 to 1v8 LDO (enabled on startup)

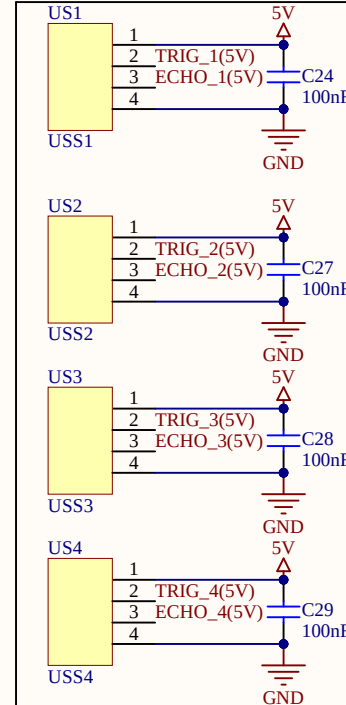


ToF/I2C Pullups

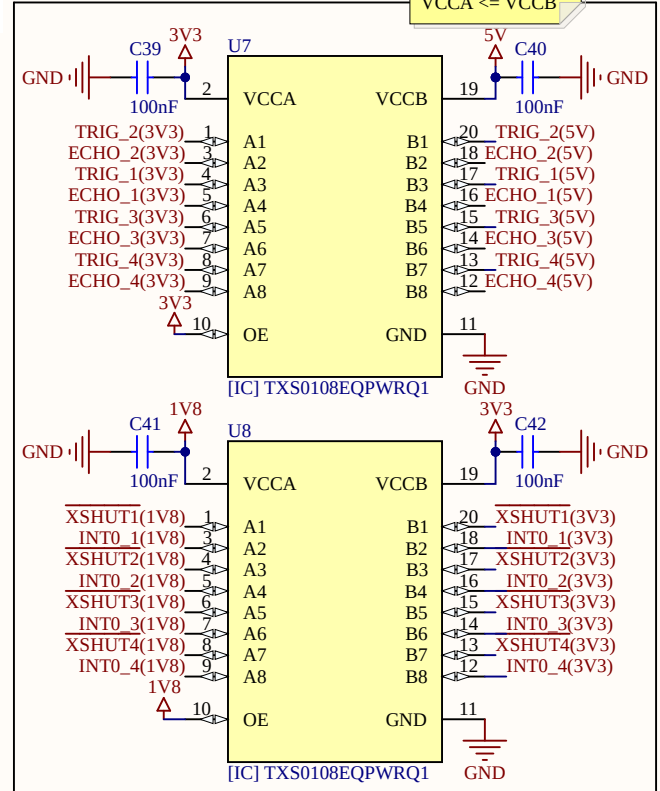


Note: GPIO1/INT0 and XSHUT pins for the ToF are pulled-up on the ToF breakout board

Headers for ultrasonic sensors

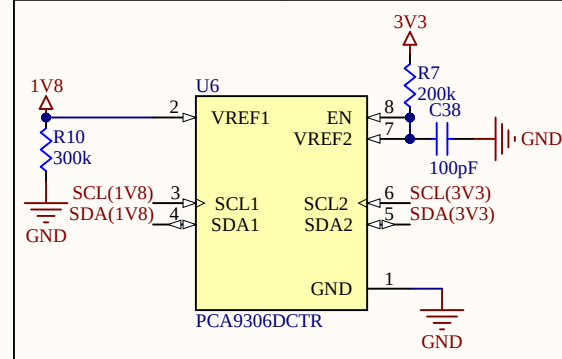


Bidirectional level shifters for distance sensors



B

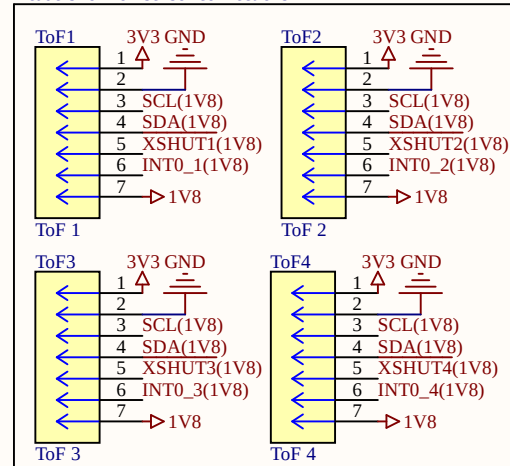
I2C level-shifter in translator configuration



C

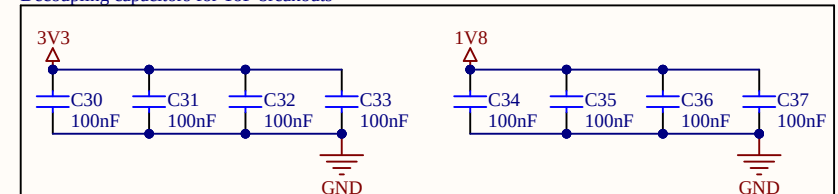
See the wiki page for the pulldown resistor calculation.

Headers for ToF sensor connections



D

Decoupling capacitors for ToF breakouts



Project/Vehicle: Distance Sensor Interface

Author(s):
-Aiden Shepler
-*
-*
-*

Revisor(s):
-*
-*
-*
-*

MIL

Level-shifting circuitry for ToF and headers for both ToF and ultrasonic sensors

Git Repo: *
Git Hash:



Date: 2/20/2026

Revision: 0

Size: A

File: Distance-Sensors.SchDoc

Sheet 2 of 2

A

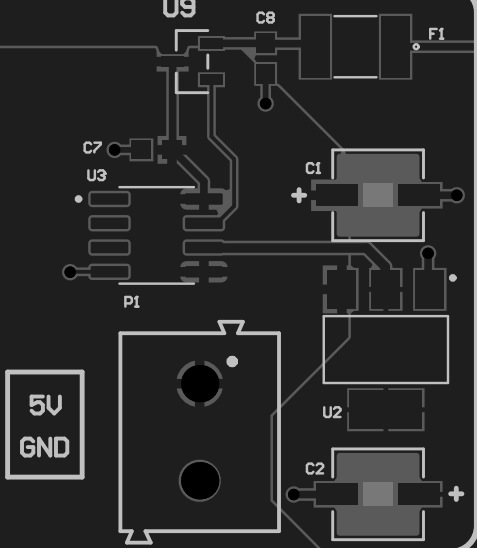
B

C

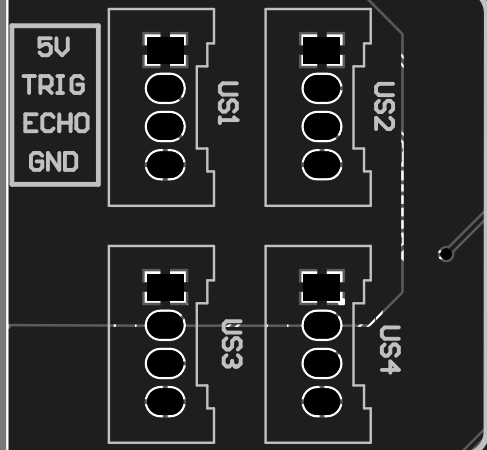
D

Distance Sensor Board [v2]
Aiden Shepler 1/25/2026

Power

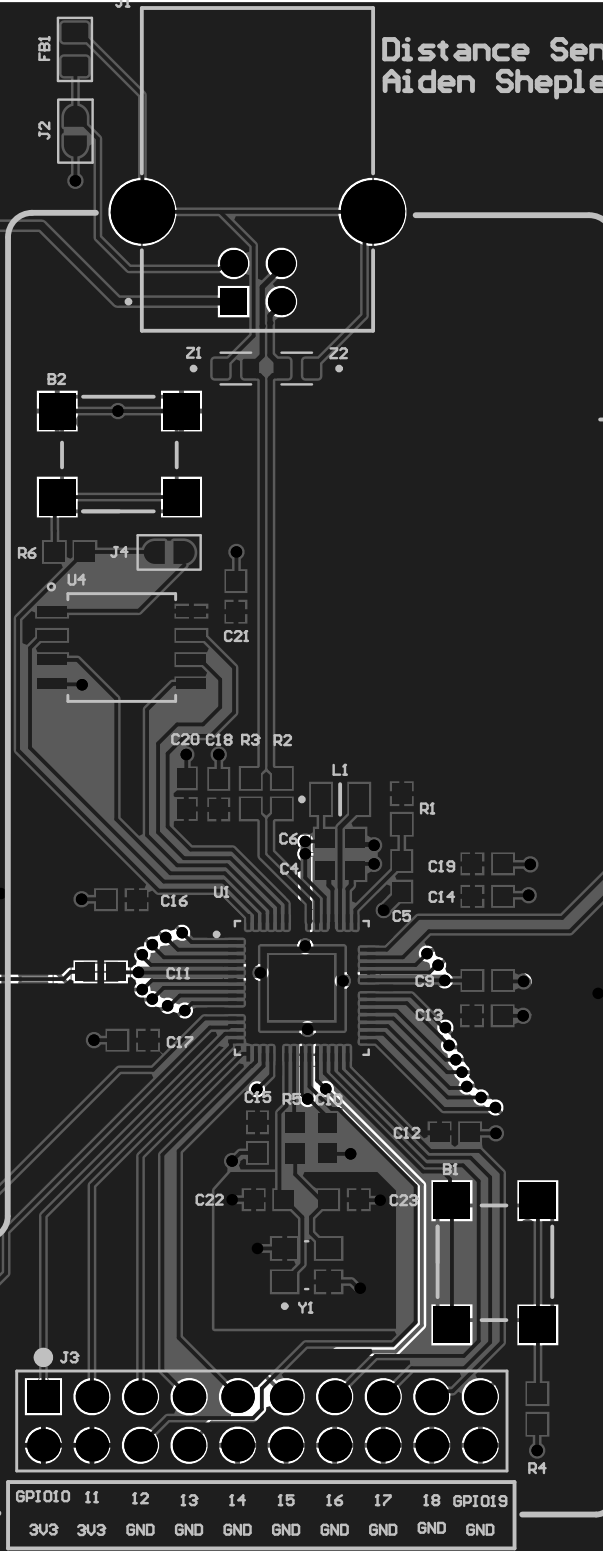


Ultrasonic Sensors

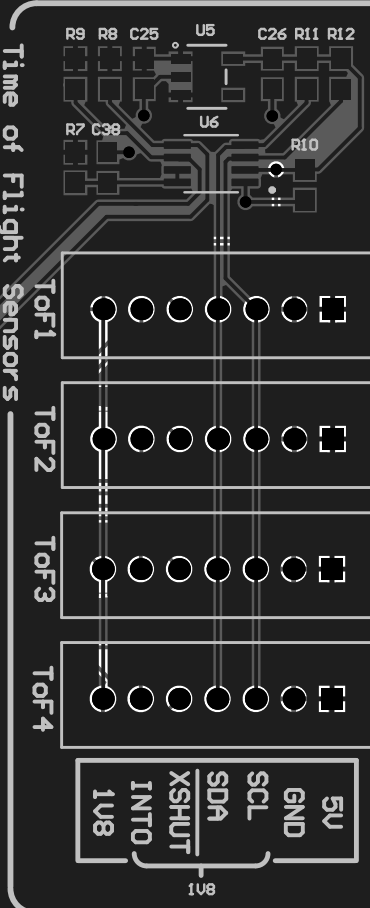


pTX
pRX
GND

MCU & Peripherals



Time of Flight Sensors



5V
GND
SCL
SDA
XSHUT
INT0
1V8

GPIO10 11 12 13 14 15 16 17 18 GPIO19
3V3 3V3 GND GND GND GND GND GND GND

Board Stack Report